

Premier League AI-Prediction

67130700346 Chaiyaporn Homtean
67130700357 Krittin Chaisuvirat
67130700361 Pawornwit Maneenet



Research Problem

Football prediction is highly complex due to multi-factorial variables such as Team Form, Market Value, Elo Ratings, and H2H stats. Traditional statistics offer limited accuracy, while fan predictions are often skewed by subjective bias

PLAYER	VALUE
Erling Haaland	£155.23m
Bukayo Saka	£129.36m
Florian Wirtz	£120.73m
Declan Rice	£103.48m
Cole Palmer	£103.48m
Alexander Isak	£103.48m
Rodri	£94.86m
Phil Foden	£86.24m
Alexis Mac Allister	£86.24m
Moises Caicedo	£77.61m

Champion (1)	
Champions League (2-4)	
Europa League (5-6)	
Europa Conference League (7)	
Higher Mid Table (8-12)	
Lower Mid Table (13-17)	
Relegation (18-20)	

380

Match/Seasons

3

Prediction Output

01

Why It Matters

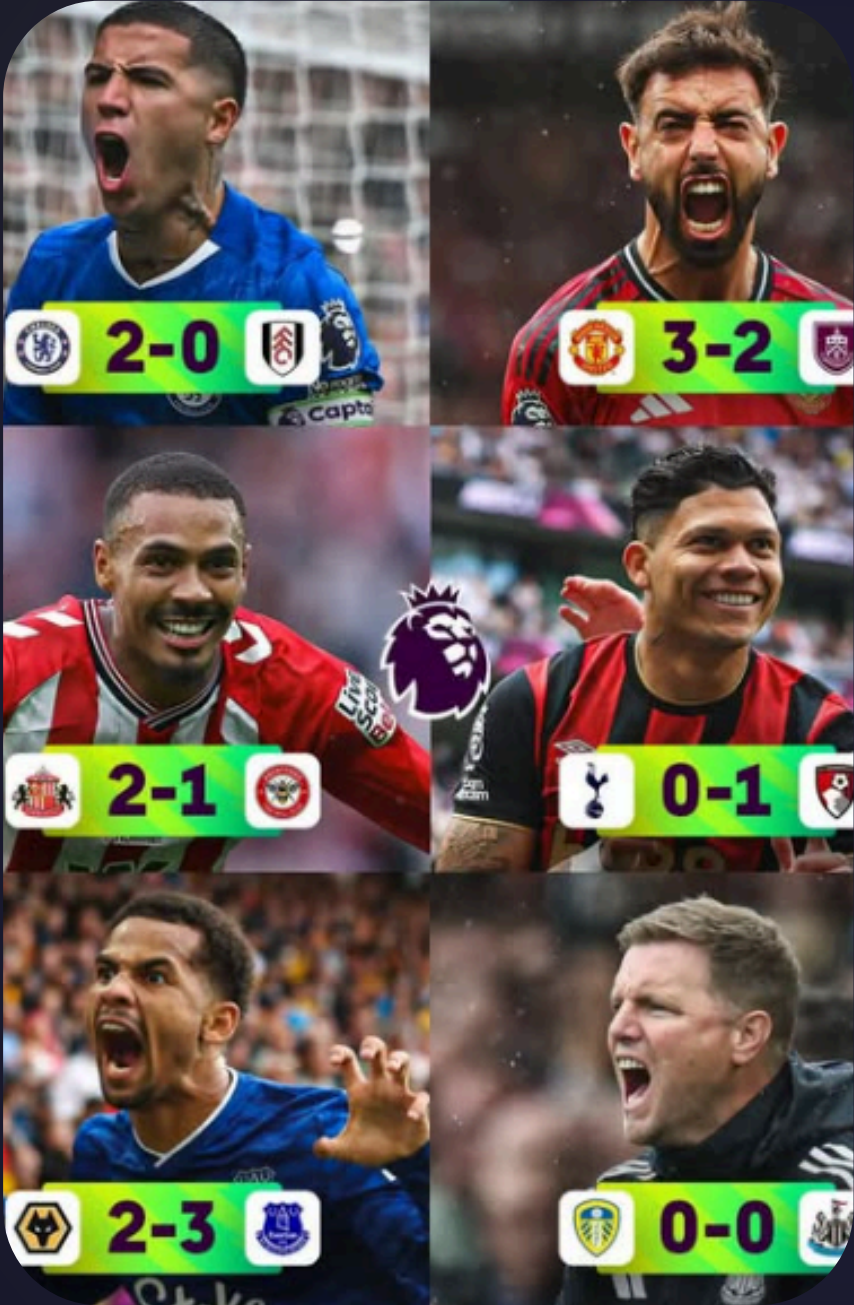
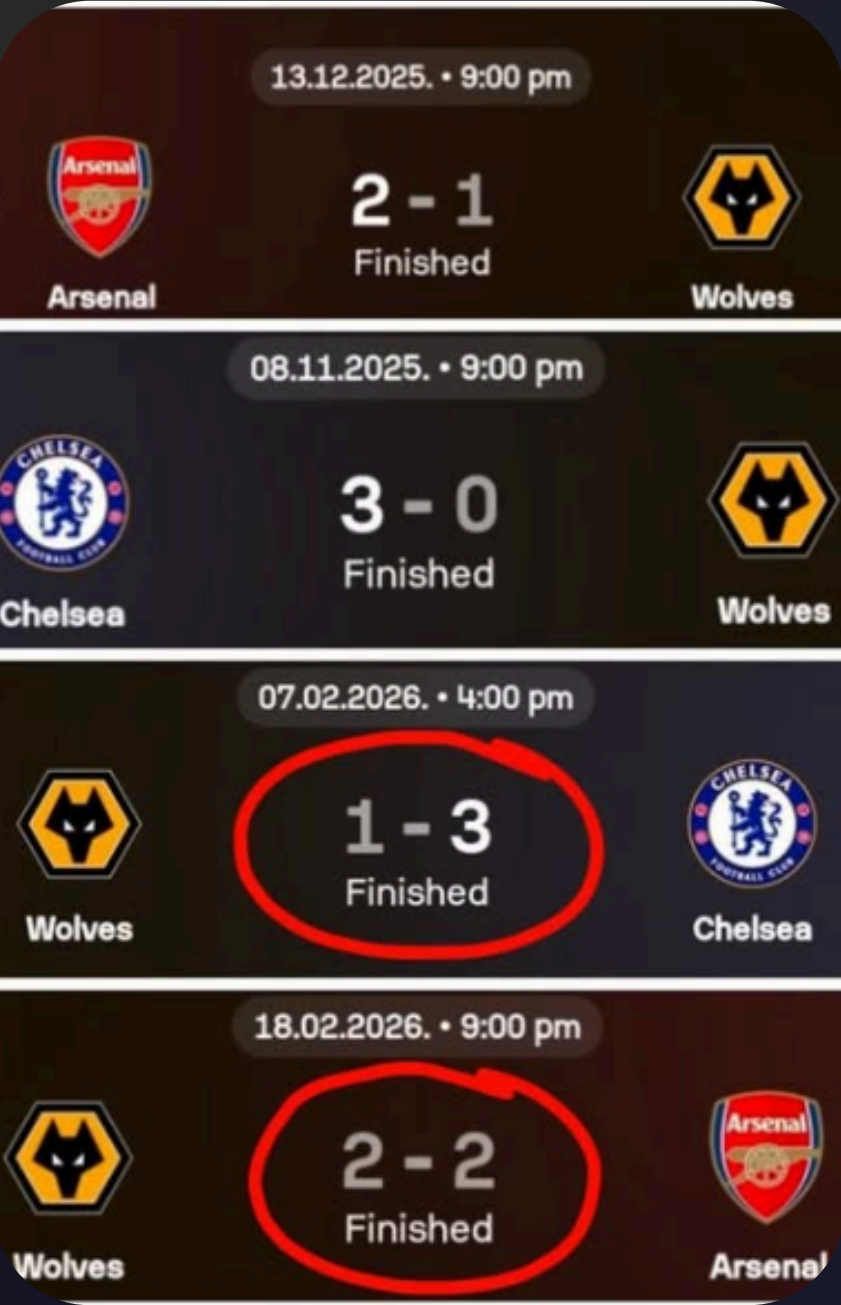
- High-Value Industry
- Data Diversity
- AI Advantage

02

Research Questions

- Stacking Ensemble vs. Single Models
- Deep Factors & Feature Importance
- AI vs. Wisdom of Crowds vs. Experts

Project Goal

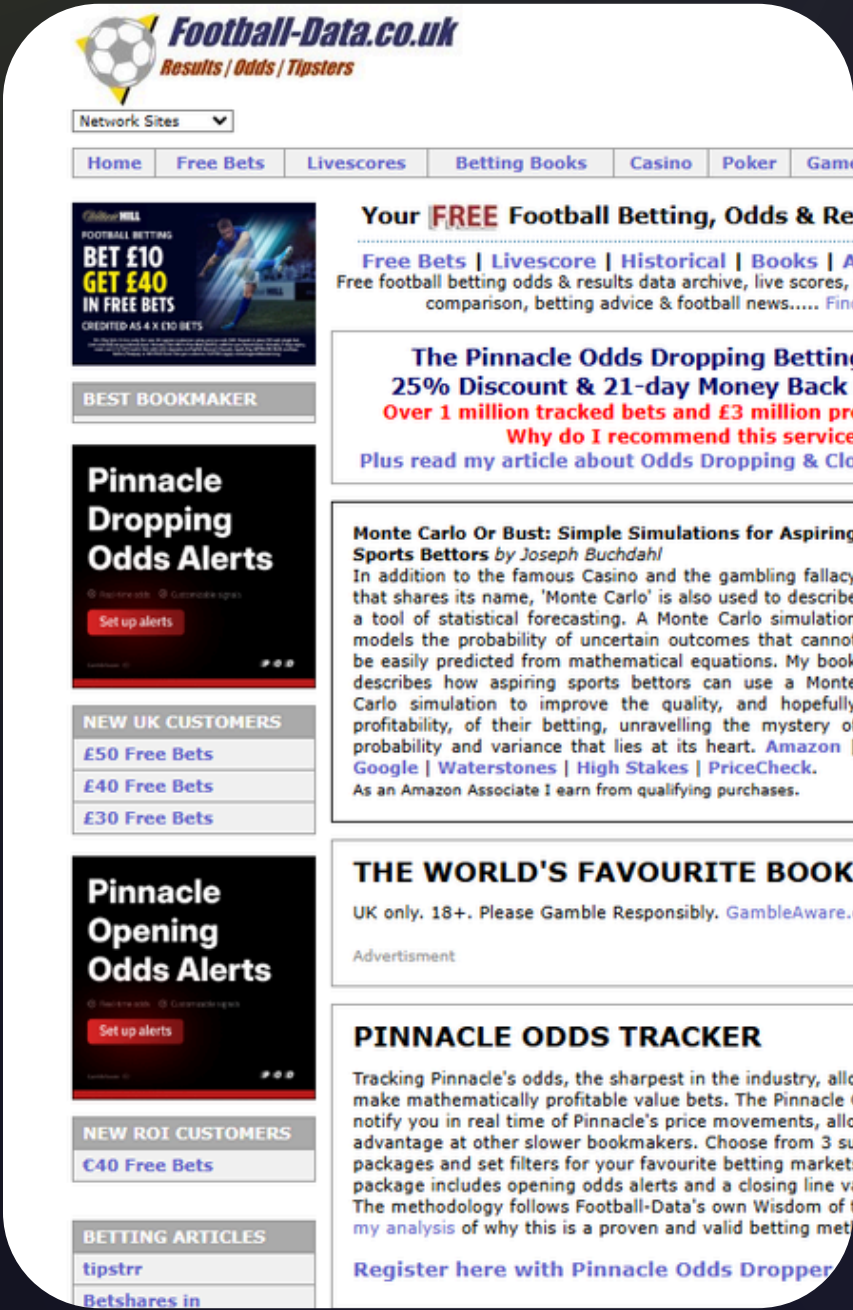


Developing an AI Model for Superior Predictive Accuracy

Create an interactive and explainable AI dashboard for predicting Premier League match results with over 55% accuracy across all three outcomes (Win/Draw/Loss)



Dataset



PL Match 2021–2025 : football-data.co.uk

Historical Match Data and In-game Statistics for the Past Five Seasons, Including Opening Betting Odds.

Market Values : transfermarkt.com

Historical Squad Value Analysis across the last five seasons to evaluate the variation in Player Quality



FPL Baselines2024 : Official FPL API

Collecting 'Fan Ownership Scores' to measure player popularity and supporter confidence, establishing a baseline for comparative analysis against AI-driven predictions

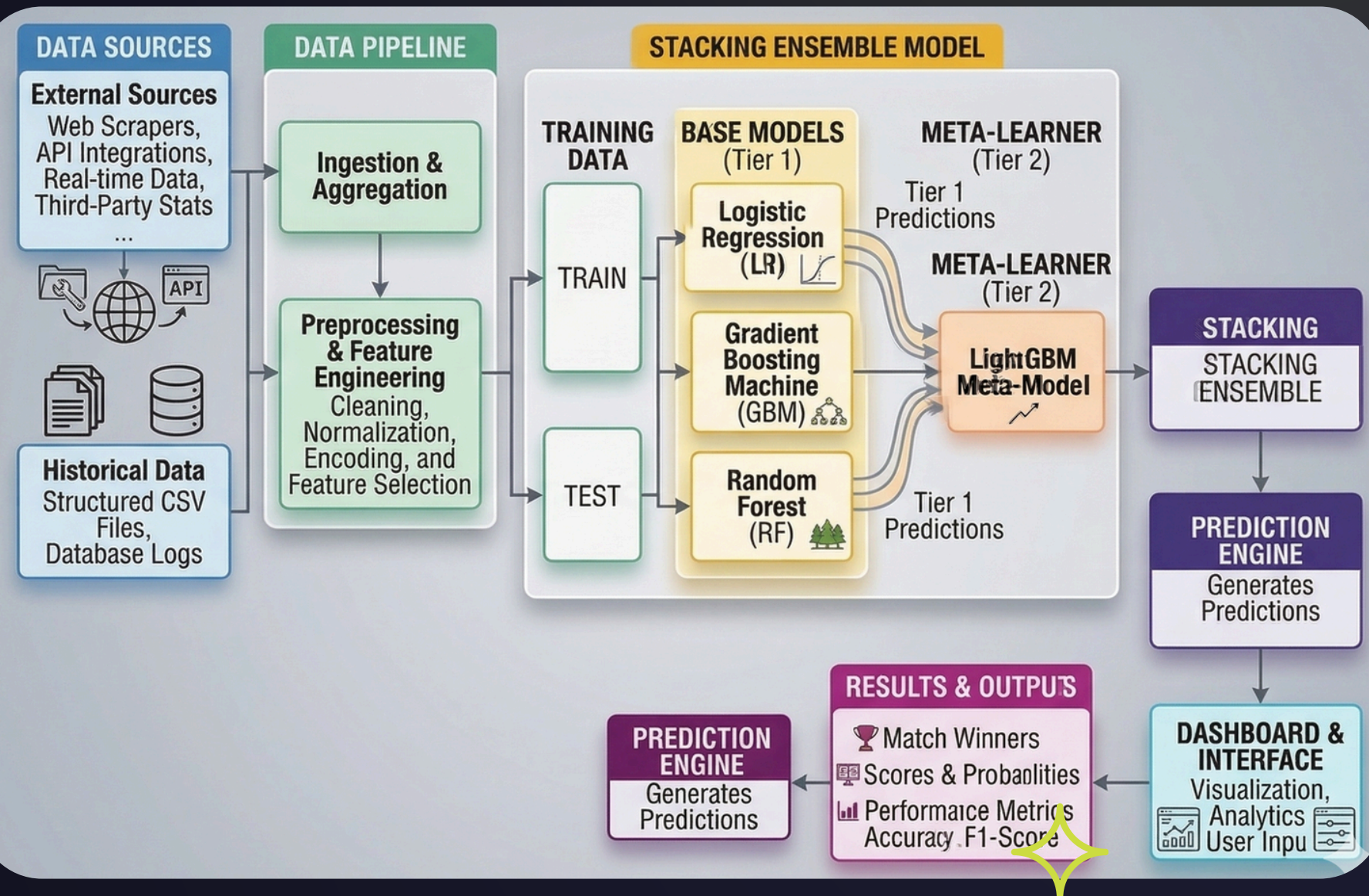
Sutton predictions and BBC

"Integrating 'Expert Insights' as a qualitative benchmark to evaluate the predictive accuracy of our AI model against industry-leading analysts



ML Methods

Random Forest Classifier



Stacking Ensemble

Base Learners :Implement an ensemble of three diverse models to train and predict outcome probabilities, where the output of each model is subsequently passed as a set of 'meta-features' for final prediction

Meta-Learner :Employ a Logistic Regression model to aggregate the outputs from the first layer, learning the optimal weights to assign to each base model for final prediction.



Methodology

Data Collection : ELO Ratings,Squad Market Value (Transfermarkt),Team Form and Goal Metrics (Form & Goals),Wisdom of Crowds

Data Partitioning: Allocating 80% for the Training Set to train the model and 20% for the Test Set for performance evaluation.

Evaluation Metrics: Utilizing a Confusion Matrix and Accuracy Score to assess the predictive precision across all match outcomes (Win/Draw/Loss)

Challenges



01

Unquantifiable Factors

Unquantifiable Factors: Football involves several unpredictable variables—such as mid-game red cards, sudden injuries to key players, referee influence, or managerial changes—that may not be fully captured or represented within historical CSV datasets



02

Class Imbalance

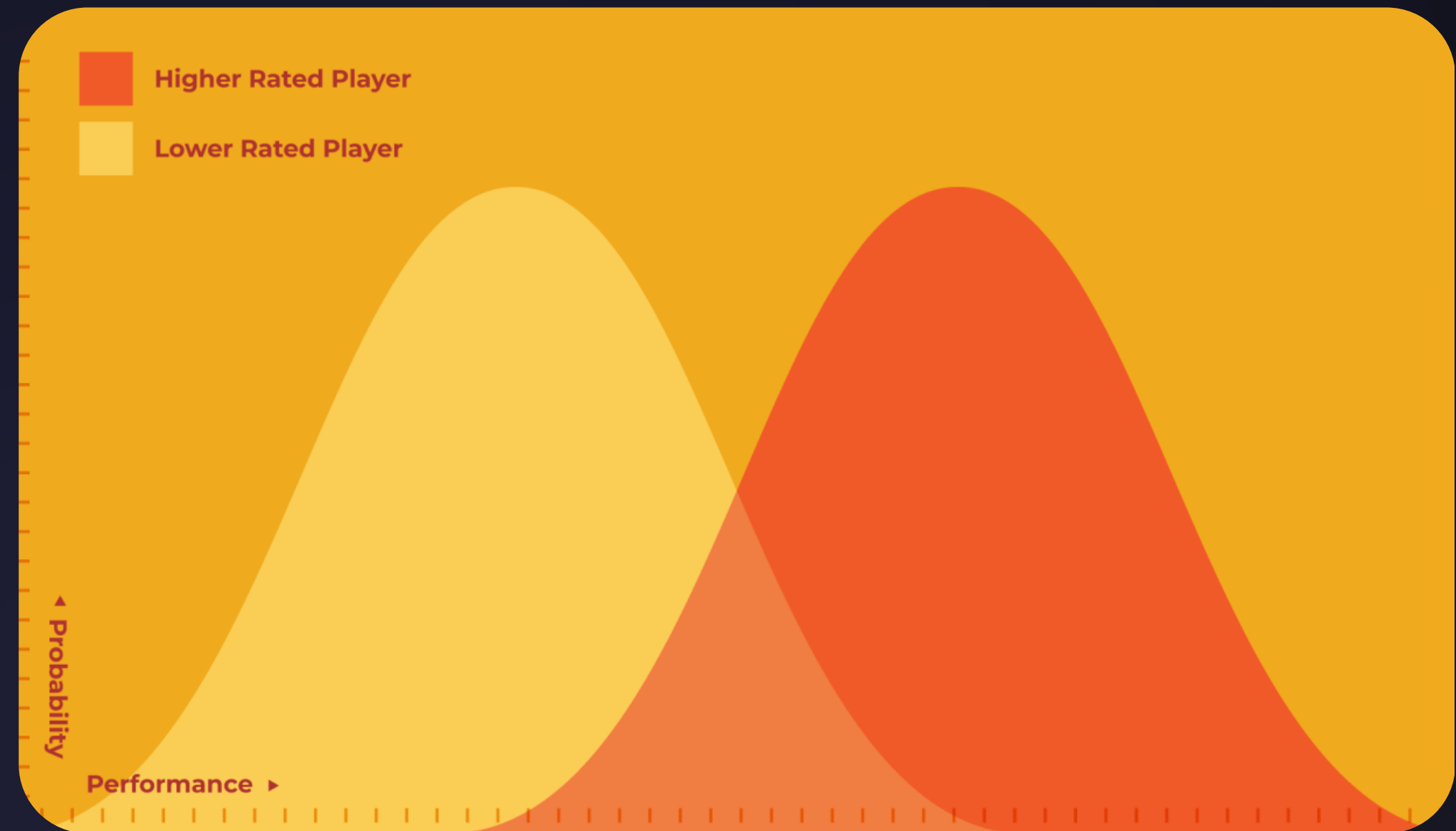
"The statistical frequency of 'Draws' is significantly lower than that of wins or losses, leading to a potential model bias toward decisive outcomes. To mitigate this class imbalance, precise hyperparameter tuning is required to improve the model's sensitivity to Draw results



Developing a Machine Learning Model Better

Elo Rating System

- Dynamic Adjustment: All teams are initialized with a baseline (e.g., 1,500 points). ✨
The points exchanged after a match depend on the Expected Outcome
- The Underdog Effect: If a lower-rated team (Underdog) defeats a higher-rated team (Favorite), they receive a significant ratings boost, while the Favorite suffers a substantial point loss.
- Predictability Weighting: Conversely, if a Favorite wins as expected, the point exchange is minimal, reflecting that the outcome provided little new information about their skill level.



The Elo rating system measures the relative strength of a player in some games, such as chess, compared to other players. Its creator, Arpad Elo, was a physics professor in the United States and a chess master who worked to improve the way the U.S. Chess Federation measured their players' skill levels.

Developing a Machine Learning Model Better

Last 5 Matches Performance

- Form L5 serves as a critical indicator of a team's 'Current Momentum.' It quantifies recent performance by aggregating the points earned over the most recent five matches ✨
- Calculation Methodology
To provide a consistent numerical input for the Random Forest model, we apply a standard points-based system:
Win (W): 3 Points ,Draw (D): 1 Point ,Loss (L): 0 Points



Developing a Machine Learning Model Better

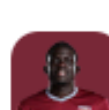
Goal Statistics (Avg. GF & Avg. GA)

- Average Goals For (Avg. GF): The mean number of goals scored by a team over their last 5 matches.
 - Reflects: Offensive Potency
- Average Goals Against (Avg. GA): The mean number of goals conceded by a team over their last 5 matches.
 - Reflects: Defensive Solidity

Goals >

1		Erling Haaland Manchester City	22
2		Igor Thiago Brentford	19
3		Antoine Semenyo Manchester City	15
4		João Pedro Chelsea	14
5		Danny Welbeck Brighton and Hove Albion	12
6		Hugo Ekitiké Liverpool	11
6		Viktor Gyökeres Arsenal	11
8		Dominic Calvert-Lewin Leeds United	10
8		Harry Wilson Fulham	10
10		Junior Kroupi Bournemouth	9

Assists >

1		Bruno Fernandes Manchester United	16
2		Rayan Cherki Manchester City	8
3		Erling Haaland Manchester City	7
4		James Garner Everton	6
4		Jarrod Bowen West Ham United	6
4		Harry Wilson Fulham	6
4		Mohamed Salah Liverpool	6
4		Jack Grealish Everton	6
9		Granit Xhaka Sunderland	5
9		El Hadji Malick Diouf West Ham United	5

Total Passes >

1		Virgil van Dijk Liverpool	
2		Lewis Dunk Brighton and Hove Albion	
3		Trevoh Chalobah Chelsea	
4		Joachim Andersen Fulham	
5		Elliot Anderson Nottingham Forest	
6		Jan Paul van den Broek Brighton and Hove Albion	
7		Marcos Senise Bournemouth	
8		Ezri Konsa Aston Villa	
9		Dominik Szoboszlai Liverpool	
10		Declan Rice Arsenal	

Developing a Machine Learning Model Better

Squad Market Value

- Data Source: Market valuations are aggregated from Transfermarkt (e.g., Manchester City valued at approximately €1.0 Billion) ✨
- Feature Engineering: We calculate the Market Value Differential (MV_Diff) between competing squads as a key predictive input.
- Analytical Rationale: Over the long term, higher-valued squads tend to exhibit greater Performance Consistency. By incorporating this metric, the model can significantly mitigate "Stochastic Noise" (unpredictable variance), leading to more stable and reliable predictions.

MARKET VALUE UPDATE SQUAD VALUES				
		UPGRADE / DOWNGRADE		NEW SQUAD VALUE
	MANCHESTER CITY	-3.9%	-€41.5M	€1.036M
	MANCHESTER UNITED	-4.4%	-€40.3M	€879.3M
	CHELSEA	-1.5%	-€13.0M	€878.5M
	LIVERPOOL	-0.6%	-€5.5M	€861.0M
	TOTTENHAM	-7.7%	-€54.0M	€643.3M
	ARSENAL	-0.3%	-€1.5M	€562.0M
	LEICESTER CITY	-5.9%	-€32.3M	€515.8M
	ASTON VILLA	+0.3%	+€1.2M	€414.3M
	EVERTON	-3.5%	-€15.0M	€413.8M
	WOLVERHAMPTON	-4.9%	-€18.8M	€365.3M
	WEST HAM UNITED	-1.1%	-€4.0M	€350.3M
	LEEDS UNITED	+2.8%	+€7.3M	€266.0M
	BRIGHTON	-2.1%	-€5.5M	€251.5M
	SOUTHAMPTON	-3.1%	-€7.9M	€247.6M
	CRYSTAL PALACE	-3.8%	-€9.5M	€240.0M
	NEWCASTLE UNITED	-9.9%	-€24.5M	€220.9M

Developing a Machine Learning Model Better

H2H

- Head-to-Head (H2H) statistics analyze the last five encounters between two specific teams. This feature identifies historical dominance or 'Bogey Team' scenarios, where a statistically stronger side consistently struggles against a specific opponent regardless of overall league standing.

	England แมนเชสเตอร์ ซิตี้		England เชลซี
แต้มต่อเกม 2.03		แต้มต่อเกม 1.55	
5 นัดล่าสุด W L D L W		5 นัดล่าสุด L L L L W	
ภาพรวม เหย้า เยือน		ภาพรวม เหย้า เยือน	

สถิติโดยรวม	ภาพรวมฤดูกาล	สถิติโดยรวม
30	แข่งแล้ว	31
18	ชนะ	13
7	เสมอ	9
5	แพ้	9
2	อันดับในลีก	6

สถิติโดยรวม	เฉลี่ยฤดูกาล	สถิติโดยรวม
2.03	แต้มต่อนัด	1.55
2.93	ประตูเฉลี่ย	2.94
2	ได้ประตูเฉลี่ย	1.71
0.93	เสียประตูเฉลี่ย	1.23

สถิติโดยรวม	เกมรุก / เกมรับ	สถิติโดยรวม
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Model Tuning

v1.0 (44.8%)

Step 1: Baseline Model (Logistic/DT)

Problem: The model exhibits high variance and demonstrates poor generalization when tested on different seasons, leading to a significant drop in predictive accuracy.

Development Strategy: Implement a long-term ELO rating system to better capture relative team strength and apply Feature Scaling to normalize input variables for improved model stability.

v2.5 (51.2%)

Step 2: Feature Engineering & Imbalance

Surpassing Expert Baseline (47.1%)

Problem: The model fails to accurately predict Draw outcomes and currently underperforms compared to 'Fan Confidence' metrics, which maintain a higher average accuracy.

Development Strategy: Incorporate 'Squad Market Value' (Transfermarkt) as a key feature and implement Class Weight adjustments to specifically enhance the model's sensitivity toward predicting draws.



Model Tuning

v3.8 (53.9%)

Step 3: Advanced Single Model (XGBoost)

Underperforming Public Sentiment (54.7%)

Problem: Individual models have reached their performance ceiling and are currently unable to aggregate diverse data streams effectively enough to outperform 'Wisdom of Crowds' confidence levels.

Development Strategy: Initiate Stacking Ensemble experiments to synthesize the unique strengths of multiple models, creating a more robust and comprehensive predictive output.

v5.0(55.3%)

Step 4: Stacking Ensemble (Meta-Mode)

Outperforming Experts and Public Sentiment (+0.5%)

Problem: The predicted probabilities are poorly calibrated, meaning the model's confidence levels do not consistently align with actual observed outcomes.

Development Strategy: Implement a Stacking Classifier integrated with a Logistic Meta-Model and apply Probability Calibration techniques to ensure predictive reliability until all performance benchmarks are surpassed.



Experimental Results



CHAMPION

 **ARSENAL**

TOP 4

 **ARSENAL**  **MAN CITY**  **LIVERPOOL**  **MAN UNITED**

RELEGATION

 **WOLVES**  **LEEDS**  **BURNLEY**

SURPRISE PACKAGE

 **EVERTON**

TOP SCORER

 **HAALAND**

BEST SIGNING

 **ZUBIMENDI**

Key Findings

Findings indicate that the Stacking Ensemble architecture consistently outperforms individual models, delivering a 2–3% improvement in overall accuracy.

The ELO rating differential was found to be the most influential predictive factor, followed by recent form (L5 window) and squad market value.

55.3%

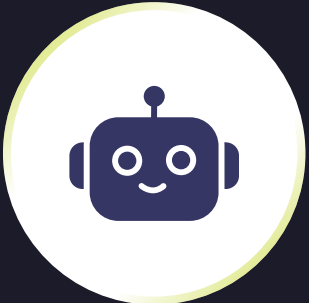
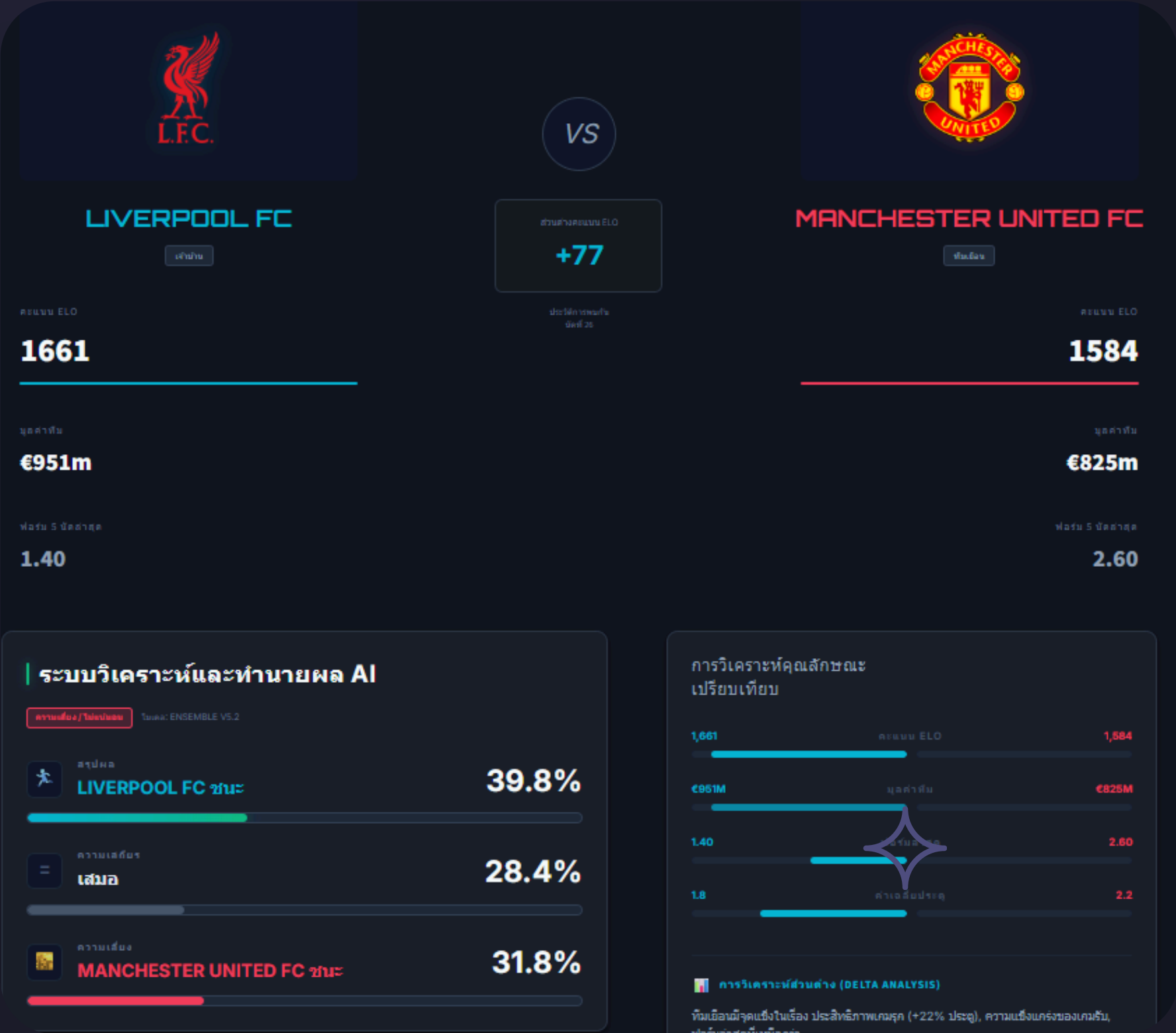
Overall Accuracy

+0.5%

Exceeding the Wisdom of
Crowds Baseline



Dashboard Showcase



Tech Stack

Python · Streamlit
scikit-learn · XGBoost
pandas · Altair
Hugging Face Spaces



Key Dashboard Features

Head-to-Head (H2H) Statistical Comparison
AI-Driven Probability Calculator
Comparative Feature Analysis
Calibration Plot: AI Confidence vs. Fan Sentiment



References



Academic & Technical Inspiration

Elo, A. E. (1978). "The Rating of Chessplayers, Past and Present". Arco Pub. (This work provides the fundamental algorithm for calculating skill levels in competitive teams or individuals, known as the Elo rating system)



Data Sources

- Football-Data.co.uk: Historical Match Results, In-game Statistics, and Betting Odds.
- Transfermarkt: Professional Player & Squad Market Values.
- Official Fantasy Premier League (FPL) API: Fan Ownership Data & Team Strength Indices.
- The Guardian Open Platform: Sports News & Match Preview Content (Sentiment Analysis).



Thank You For Attention

